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Uncovering novel scientific insights with a synergistic GNN-LLM framework

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ABSTRACT

The exponential growth of scientific literature demands intelligent systems capable of uncovering emerging knowledge associations and fostering creativity. While graph neural networks (GNNs) excel at modeling literature structures, their static temporal modeling and lack of semantic awareness limit the discovery of interpretable signals. Conversely, large language models (LLMs) offer deep semantic reasoning but struggle to find non-obvious, structurally-grounded patterns without structured input. To address these limitations, this paper proposes a multi-stage GNN-LLM framework that integrates structural pattern recognition and semantic interpretation for scientific knowledge discovery. The framework begins with a Semantic-Enhanced Temporal Graph Network (SE-TGN), which embeds paper-level semantic information into an event-based temporal GNN to identify emerging keyword associations. These structurally grounded candidates are refined through the Contextual Re-ranking and Evaluation Framework (CREF), which leverages LLM capabilities to assess contextual novelty and relevance. Finally, the Generative Interpretation and Contextualization (GIC) produces human-readable explanations and research prompts to support innovation. Experiments in two scientific domains demonstrate the effectiveness of the framework in discovering semantically rich, contextually grounded, and forward-looking knowledge associations, illustrating its potential to support interpretable and creativity-driven scientific exploration.

1. Introduction

The accelerating growth of scientific publications has dramatically increased the complexity of discovering novel and meaningful knowledge associations [1–3]. Researchers must now process large-scale, heterogeneous, and rapidly evolving literature to stay abreast of emerging themes, detect latent connections, and formulate new research questions [4]. Manual review and keyword-based search techniques are increasingly insufficient in this environment, creating a pressing need for intelligent systems that can detect structural signals and support creative scientific inquiry.

Large Language Models (LLMs) have emerged as powerful tools for language understanding, semantic synthesis, and text generation [5]. Leveraging pretraining on large corpora, LLMs excel at summarization, scientific question answering, and conceptual explanation [6–8]. However, when tasked with proactive knowledge discovery, LLMs face important limitations. Without structured input or targeted context, they often produce outputs that are generic, speculative, or disconnected from the underlying data [9–11]. Furthermore, their sequential architecture is not naturally suited to model complex relational structures or the temporal dynamics inherent in scientific evolution [12].

Graph Neural Networks (GNNs), in contrast, offer strong capabilities in structural pattern learning [13]. When applied to scientific graphs such as citation networks or keyword co-occurrence graphs, GNNs can learn topological representations and predict missing or future links. Models like GCN, GraphSAGE, and GAT have demonstrated success in capturing local and global graph properties through neighborhood aggregation [14–16]. Nonetheless, existing GNN-based literature mining approaches face notable limitations: (1) most operate on static snapshots and lack temporal granularity, hindering their ability to track concept drift and emergent associations [17]; (2) they often omit the semantic content underlying relationships, such as the thematic context of a paper where a keyword pair occurs [18]; and (3) they typically lack mechanisms for semantic interpretation or explanation, reducing their value in hypothesis generation or research ideation [19].

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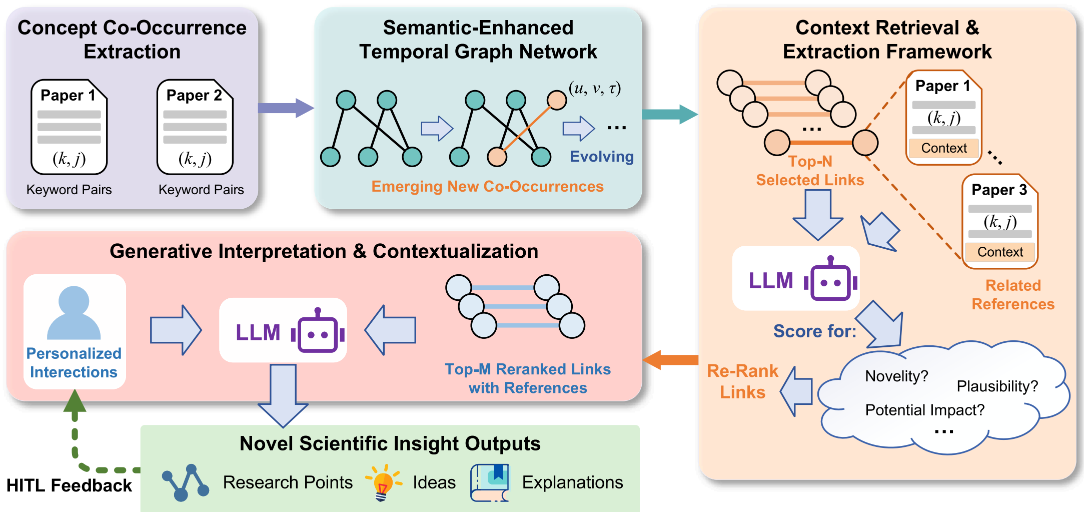


Fig. 1. Overview of the proposed GNN-LLM framework. This multi-stage pipeline integrates: 1) Concept Co-Occurrence Extraction from papers; 2) a Semantic-Enhanced Temporal Graph Network (SE-TGN) to identify emerging keyword associations; 3) a Contextual Re-ranking and Evaluation Framework (CREF) where an LLM assesses novelty and impact; and 4) a Generative Interpretation and Contextualization (GIC) module where an LLM produces human-readable explanations and research ideas. The framework synergistically combines structural pattern recognition with semantic interpretation to support scientific knowledge discovery and creativity-driven exploration.

These observations suggest a fundamental gap: while GNNs provide structural insight and LLMs offer semantic reasoning, few existing systems combine both to enable interpretable and forward-looking scientific discovery. Bridging this gap requires an integrated architecture that couples graph-based structure learning with context-aware language-based evaluation and generation.

To address this challenge, a multi-stage GNN-LLM framework is introduced that unifies temporal graph modeling, semantic evaluation, and research-oriented explanation. The first stage employs a Semantic-Enhanced Temporal Graph Networ (SE-TGN), which builds on the Temporal Graph Network paradigm by incorporating paper-level semantic embeddings into an event-driven message-passing architecture. This enables modeling of evolving keyword interactions with both topological and semantic awareness. The output of SE-TGN is a ranked list of structurally grounded, temporally emergent candidate keyword associations. In the second stage, the Contextual Re-ranking and Evaluation Framework (CREF) uses an LLM to reassess these candidates based on their novelty, contextual relevance, and potential research value. Rather than relying solely on graph structure, CREF leverages the surrounding paper context and recent usage patterns to prioritize associations that are not only likely, but meaningful. In the third stage, Generative Interpretation and Contextualization (GIC) transforms top-ranked candidates into natural language outputs–providing human-readable explanations and proposing research directions that are grounded in the predicted patterns.

This GNN-LLM synergy enables a discovery process that is both structured and semantically enriched. By tightly integrating predictive modeling with interpretive reasoning, the framework supports a form of machine-assisted scientific discovery that moves beyond link prediction toward insight generation. The overall framework is illustrated in Fig. 1. The primary contributions of this work are summarized as follows:

1. A multi-stage GNN-LLM framework is proposed that integrates temporal graph modeling and semantic language reasoning to support scientific knowledge discovery and ideation.
2. A Semantic-Enhanced Temporal Graph Network (SE-TGN) is developed to incorporate paper-level semantic embeddings into event-based link prediction, enabling structurally and semantically grounded candidate generation.
3. Two LLM-based modules–CREF and GIC–are introduced to perform contextual evaluation and explanation of GNN outputs, facilitating human-readable, insight-rich scientific guidance.

The remainder of this paper is organized as follows. Section 2 reviews existing work on graph-based knowledge discovery and recent efforts to integrate LLMs with structured data. Section 3 introduces the proposed multi-stage GNN-LLM framework, detailing the design of SE-TGN, CREF, and GIC. Section 4 describes the experimental setup, datasets, baselines, and evaluation metrics, followed by quantitative and qualitative results. Section 5 discusses the implications, limitations, and potential extensions of the framework. Finally, Section 6 concludes the paper and outlines future directions.

2. Related work

2.1. Graph-based approaches for dynamic link prediction in scientific knowledge graphs

Network-based modeling of scientific literature, particularly via keyword co-occurrence [20], citation [21], or author collaboration graphs [22], has long served as a foundation for understanding thematic structures and discovering emerging research areas [23]. Early link prediction methods, such as Common Neighbors, Jaccard similarity, and Adamic-Adar index, provided efficient heuristics for inferring latent connections [24–26]. However, these methods rely purely on local topology and lack the expressive capacity to capture high-order patterns or incorporate node- and edge-level attributes, limiting their utility in dynamic and semantically rich knowledge domains [27].

Graph neural networks have significantly advanced the field by learning task-specific node and edge representations through iterative message passing [28]. Models such as Graph Convolutional Networks (GCNs) [14], GraphSAGE [15], and Graph Attention Networks (GATs) [16] have been applied to scientific graphs to predict keyword associations, cluster topics, and detect emergent research communities. These methods are typically evaluated on static graph snapshots, representing an aggregated view of past co-occurrence or citation patterns [29,30]. While effective in many settings, such static models inherently ignore the temporal evolution of scientific knowledge–an aspect that is critical for anticipating emerging trends [31].

To address temporal dynamics, event-based and continuous-time graph neural network models have been developed [32,33]. For example, TGNs process sequences of time-stamped interactions, updating node memories through temporal encoding and message passing mechanisms [34,35]. This allows them to capture recency-sensitive and context-aware structural patterns. Recent adaptations have explored incorporating auxiliary semantic features, such as document embeddings from the papers that induce keyword co-occurrences, into the GNN message passing pipeline. These enhancements improve the model's ability to surface not only structurally probable, but also semantically meaningful future associations. Our proposed Semantic-Enhanced Temporal Graph Network (SE-TGN) builds on this line of work, embedding paper-level semantics into an event-driven temporal GNN for precise, context-aware candidate discovery.

2.2. Large language models and integration with structured knowledge

Recently, large language models, such as GPT [5], LLaMA [36], or the emerging DeepSeek [37], have demonstrated substantial capabilities in natural language understanding and generation. In the scientific domain, their use is expanding to tasks such as text summarization [7], concept explanation [8], and question answering [38]. These models exhibit strong abilities in capturing semantic relationships and producing contextually coherent text, positioning them as useful tools for assisting research interpretation and communication.

Despite these advances, LLMs remain limited in their ability to proactively discover novel, structurally-defined associations from unstructured scientific corpora [39]. Without guidance from structured context, LLM outputs often lack grounding or relevance, and may fail to reflect the evolving dynamics of research domains [9,40,41]. Furthermore, their sequential token-based architecture makes them inherently less suited for tasks involving temporal reasoning or relational graph analysis, which are critical for uncovering emerging scientific insights [42,43].

To address these gaps, a growing body of work has begun exploring the integration of LLMs with graph-based methods [44–46]. Early efforts have focused on enriching graph node representations using LLM-generated features or feeding structured graph outputs into LLMs to support downstream reasoning tasks [47,48]. While these approaches highlight the potential for synergy, many operate on static graph snapshots and lack the temporal granularity necessary to model the evolution of knowledge [49,50]. Our work distinguishes itself by integrating LLMs with a temporal graph network. This allows our framework not just to reason over structured knowledge, but to specifically identify and evaluate emerging associations in a dynamic, event-driven context, thus bridging the gap between static knowledge representation and forward-looking scientific discovery.

The framework presented in this paper builds on this emerging direction by combining temporally-aware GNNs with LLM-based semantic reasoning in a multi-stage architecture. The integration enables structurally grounded candidate generation, contextual relevance assessment, and interpretive insight generation–bridging the gap between structural prediction and innovation-oriented scientific discovery.

3. Methodology

The overall architecture is composed of three core modules: a Semantic-Enhanced Temporal Graph Network (SE-TGN) for predictive modeling, a Contextual Re-ranking and Evaluation Framework (CREF) for semantic value assessment, and a Generative Interpretation and Contextualization (GIC) module for research insight generation. Supporting components such as task formulation, data preprocessing, and optional human-in-the-loop (HITL) mechanisms ensure a complete and extensible pipeline.

3.1. Task formulation

This work formulates scientific knowledge service as a multi-stage process that begins with predictive modeling and culminates in human-readable insight generation. The goal is not only to identify potential emerging concept associations in scientific literature, but also to semantically evaluate their research value and articulate them in a form that supports idea generation and hypothesis exploration.

Given a corpus of timestamped scientific publications $\mathcal{P} = \{p_1, \dots, p_N\}$, each paper is represented by a publication time τ_i, a set of standardized concepts $K_i \subseteq V$ from a global vocabulary V, and a textual field $Text_i$ consisting of its title and abstract. From this corpus, a temporal stream of concept co-occurrence events is derived, where each event $e_j = (u_j, v_j, \mathrm{pid}_j, \tau_j)$ captures the co-occurrence of two distinct concepts u_j and v_j in paper pid_j at time τ_j. The full sequence $\mathcal{S} = \{e_1, \dots, e_M\}$ serves as the temporal substrate for graph-based modeling.

To incorporate semantic information, the textual content of each paper is encoded into a fixed-length vector using a pretrained sentence encoder:

$$\mathbf{m}_{\mathrm{pid}_j} = f_{\mathrm{enc}}(Text_{\mathrm{pid}_j}). \tag{1}$$

Based on all events observed up to time T, the system aims to surface previously unobserved concept pairs that are likely to co-occur in the near future, i.e., within a target window $[T, T+\Delta]$. However, the task extends beyond prediction. For each predicted candidate pair (u, v), the system further conducts contextual evaluation of its novelty, plausibility, and potential research value, followed by generation of a concise and interpretable explanation that may include relevant definitions, rationale for their connection, and example research questions or hypotheses. These steps collectively constitute a structured scientific knowledge service.

The final output of the system is a set of tuples, where each element combines structural prediction, semantic assessment, and insight generation:

$$\mathcal{O} = \left\{ \left((u, v),\ P(u, v),\ S_{\mathrm{value}}(u, v),\ \mathcal{R}(u, v) \right) \right\}. \tag{2}$$

Here, $P(u, v)$ denotes the predicted co-occurrence probability, $S_{\mathrm{value}}(u, v)$ represents the semantic evaluation score, and $\mathcal{R}(u, v)$ is the generated explanatory output, serving as a knowledge-guided interface for researchers.

3.2. Data preparation and framework overview

The proposed framework operates on a corpus of timestamped scientific publications. Each paper is associated with a publication time, a title and abstract, and a set of normalized keywords drawn from a controlled concept vocabulary. From this corpus, a stream of temporal concept co-occurrence events is constructed. For each paper, all unordered pairs of distinct keywords are extracted to form events of the form $e_j = (u_j, v_j, \mathrm{pid}_j, \tau_j)$, where τ_j is the publication timestamp. These events are chronologically ordered and form the evolving conceptual graph input.

To incorporate semantic signals, the textual content of each paper (title and abstract) is encoded into a dense vector $\mathbf{m}_{\mathrm{pid}_j}$ using a pretrained sentence encoder such as SciBERT or Sentence-BERT. These semantic vectors are later injected into the graph model as dynamic features on temporal edges.

The overall system follows a three-stage pipeline. First, the SE-TGN models the temporal evolution of concept relationships using both topological and semantic information, and predicts future keyword associations based on link likelihood. The resulting candidate pairs are passed to the CREF, which invokes a large language model to assess their research value through semantic analysis of related paper contexts. Finally, the GIC module produces natural language outputs for top-ranked candidates, offering explanations and actionable research prompts. Together, these stages form an end-to-end system that integrates structural forecasting with semantic evaluation to support machine-assisted scientific discovery.

3.3. Emerging association prediction via semantically-enhanced temporal graph modeling

The Emerging Association Prediction Module (EAPM) is responsible for detecting structurally and temporally grounded keyword associations. At its core is the SE-TGN, which extends standard temporal graph learning with document-level semantic embeddings. The input consists of a time-ordered stream of keyword co-occurrence observations $\mathcal{S} = \{o_j = (u_j, v_j, \mathrm{pid}_j, \tau_j)\}_{j=1}^{M}$, where u_j, v_j are distinct concepts (keywords), pid_j is the publication ID in which they co-occur, and τ_j is the publication timestamp.

Each observation corresponds to a directed interaction in a dynamic concept graph G_τ, where nodes represent keywords, and edges are induced by their co-occurrence in literature. For every publication, its abstract and title are encoded into a semantic embedding vector $\mathbf{m}_{\mathrm{pid}_j} \in \mathbb{R}^{d_m}$ using a pre-trained language encoder (e.g., MiniLM [51] or SciBERT [52]). These embeddings serve as dynamic edge features, enriching message passing with document-level semantics. Following the TGN architecture, each node i maintains a memory state $\mathbf{s}_i(\tau) \in \mathbb{R}^{d_s}$ and a timestamp t_i^{last} indicating its last update. As observations arrive chronologically, the model encodes temporal context via a learnable time encoder $\Phi : \mathbb{R}^+ \to \mathbb{R}^{d_t}$, and constructs semantically-enhanced messages for both endpoints of each co-occurrence. These messages are then used to update node memories via a gated recurrent unit (GRU) Algorithms 1 and 2.

To support downstream prediction, SE-TGN computes dynamic node embeddings using a GAT-based temporal aggregator over each node's recent neighbors. For a candidate keyword pair (u, v) at query time τ_{query}, their embeddings $\mathbf{z}_u, \mathbf{z}_v \in \mathbb{R}^{d_h}$ are used to compute a predicted link score:

$$P((u,v)|\tau_{\mathrm{query}}) = \sigma\big(f_{\mathrm{pred}}(\mathbf{z}_u(\tau_{\mathrm{query}}), \mathbf{z}_v(\tau_{\mathrm{query}}))\big), \tag{3}$$

where f_{pred} is a two-layer neural predictor and σ denotes the sigmoid function.

The model is trained using binary cross-entropy loss over positive and sampled negative observations. To avoid over-wide equations, we rewrite the training objective as:

$$\mathcal{L} = -\sum_{(u,v,\tau)\in\mathcal{B}} \big[y \log P((u,v)|\tau) + (1-y)\log(1 - P((u,v)|\tau))\big], \tag{4}$$

where $\mathcal{B} = \mathcal{B}^+ \cup \mathcal{B}^-$ includes positive and negative keyword pairs, and $y \in \{0, 1\}$ indicates whether the pair was observed during the given period.

The trained SE-TGN outputs a ranked list of keyword pairs with high predicted likelihood during a future time period. These candidates, together with associated contextual papers, are passed to the LLM-driven stages for semantic re-ranking and interpretation. The training algorithm of Semantic-Enhanced Temporal Graph Network (SE-TGN) is presented in Algorithm 1.

Algorithm 1 Training of semantic-enhanced temporal graph network (SE-TGN).

Require: Temporal interaction stream $\mathcal{S}_{\mathrm{train}} = \{(u, v, \mathrm{pid}, \tau)\}$ sorted by time;
Paper semantic encoder f_{enc}; Time encoder Φ;
Message function f_{msg}; Memory update GRU; Link predictor f_{pred};
Number of training epochs E; Optimizer `opt`.
Ensure: Trained SE-TGN model parameters.
Initialize all node memories $\mathbf{s}_i \leftarrow \mathbf{0}$ and timestamps $t_i^{\mathrm{last}} \leftarrow 0$
Precompute paper embeddings: $\mathbf{m}_{\mathrm{pid}} \leftarrow f_{\mathrm{enc}}(Text_{\mathrm{pid}})$ for each paper
for epoch = 1 to E **do**
  Initialize accumulated loss $\mathcal{L} \leftarrow 0$
  for each interaction $(u, v, \mathrm{pid}, \tau)$ in $\mathcal{S}_{\mathrm{train}}$ **do**
    // Temporal encoding and semantic message generation
    $\phi_{\Delta\tau_u} \leftarrow \Phi(\tau - t_u^{\mathrm{last}})$
    $\phi_{\Delta\tau_v} \leftarrow \Phi(\tau - t_v^{\mathrm{last}})$
    $\mathbf{x}_{u\leftarrow v}^{\mathrm{in}} \leftarrow \mathrm{CONCAT}(\mathbf{s}_v, \mathbf{m}_{\mathrm{pid}}, \phi_{\Delta\tau_v})$
    $\mathbf{x}_{v\leftarrow u}^{\mathrm{in}} \leftarrow \mathrm{CONCAT}(\mathbf{s}_u, \mathbf{m}_{\mathrm{pid}}, \phi_{\Delta\tau_u})$
    $\mathbf{msg}_{u\leftarrow v} \leftarrow f_{\mathrm{msg}}(\mathbf{x}_{u\leftarrow v}^{\mathrm{in}})$
    $\mathbf{msg}_{v\leftarrow u} \leftarrow f_{\mathrm{msg}}(\mathbf{x}_{v\leftarrow u}^{\mathrm{in}})$
    // Memory update
    $\mathbf{s}_u \leftarrow \mathrm{GRU}(\mathbf{s}_u, \mathbf{msg}_{u\leftarrow v})$
    $\mathbf{s}_v \leftarrow \mathrm{GRU}(\mathbf{s}_v, \mathbf{msg}_{v\leftarrow u})$
    $t_u^{\mathrm{last}} \leftarrow \tau$, $t_v^{\mathrm{last}} \leftarrow \tau$
  end for
  // Link prediction and loss computation
  Sample mini-batch of positive pairs $\mathcal{B}^+$ and negative pairs $\mathcal{B}^-$
  for each $(u, v, \tau) \in \mathcal{B}^+ \cup \mathcal{B}^-$ **do**
    Compute embeddings $\mathbf{z}_u, \mathbf{z}_v$ using recent memories and temporal neighbors
    Compute score $s_{uv} \leftarrow f_{\mathrm{pred}}(\mathbf{z}_u, \mathbf{z}_v)$
    Label $y = 1$ if $(u, v, \tau) \in \mathcal{B}^+$ else 0
    $\mathcal{L} \leftarrow \mathcal{L} + \mathrm{BCE}(s_{uv}, y)$
  end for
  // Loss computation and training
  Predict link probabilities with current model
  Compute binary cross-entropy loss $\mathcal{L}$
  Update parameters to minimize $\mathcal{L}$
end for

3.4. Contextual re-ranking and evaluation framework (CREF)

Following the initial generation of candidate keyword pairs by the SE-TGN, the CREF utilizes a language model-driven mechanism for semantic assessment and prioritization. This module addresses the challenge that, while SE-TGN is effective at predicting candidate keyword co-occurrences, not all predicted links carry the same level of scientific value or novelty.

The input to CREF consists of the top-N keyword co-occurrence candidates, $C_{\mathrm{EAPM}} = \{(k_i, k_j)\}$, produced by SE-TGN, along with their associated prediction scores $s_{GNN}(k_i, k_j)$. For each candidate pair, a semantic context, denoted as $C(k_i, k_j)$, is constructed by retrieving relevant literature snippets. Given that many predicted links correspond to keyword pairs that have not yet co-occurred in the corpus, the context is constructed by retrieving papers that are independently related to k_i and k_j. A small set of titles and abstracts is then extracted and concatenated to form a surrogate contextual corpus, $C(k_i, k_j)$.

An LLM is then prompted to evaluate the research potential of each candidate (k_i, k_j) based on its semantic context. The evaluation encompasses four key dimensions: (i) *Novelty*, which estimates how unexplored the connection is in current research; (ii) *Potential Impact*, which reflects the possible significance of the connection for advancing theoretical or applied research; (iii) *Scientific Plausibility*, which evaluates the logical coherence of the connection within the context of existing literature; and optionally, (iv) *Interdisciplinarity*, which assesses the degree to which the connection spans multiple research domains.

For each dimension $f \in \mathcal{F}$, the LLM outputs a score $S_f(k_i, k_j) \in \{1, 2, 3, 4, 5\}$, along with a brief justification for the assigned score. The following prompt is designed to guide the model in assessing the relevance and scientific value of the keyword pair:

To derive an overall value score, a weighted aggregation is applied as follows:

$$S_{\mathrm{LLM}}(k_i, k_j) = \sum_{f\in\mathcal{F}} w_f \cdot S_f(k_i, k_j), \tag{5}$$

where w_f are dimension weights satisfying $\sum w_f = 1$. In our experiments, we assume equal importance for all dimensions and set the weights uniformly (i.e., $w_f = 0.25$ for each of the four dimensions). The list of candidates is then re-ranked by descending S_{LLM} scores, yielding the prioritized set C_{CREF}.

Prompt Example for CREF

> You are an expert in scientific knowledge evaluation. Given the following keyword pair (k_i, k_j), assess its potential research value based on the following dimensions:
> 1. Novelty: How novel or unexplored is this connection in the context of current research? Score from 1 to 5 (1 = Not novel, 5 = Highly novel).
> 2. Potential Impact: What is the potential impact of this research topic? How might it contribute to theoretical or practical advancements? Score from 1 to 5 (1 = Low impact, 5 = High impact).
> 3. Plausibility: Does this connection make sense given the current research trends? Score from 1 to 5 (1 = Improbable, 5 = Highly plausible).
> 4. Interdisciplinarity: How interdisciplinary is this topic? Does it span across multiple research domains? Score from 1 to 5 (1 = Not interdisciplinary, 5 = Highly interdisciplinary).
> Please provide a brief justification for each score.

Each entry in the output of CREF includes the candidate pair (k_i, k_j), the SE-TGN prediction score s_{GNN}, the LLM value score S_{LLM}, and a detailed set of dimension-wise evaluations with explanations. This enables downstream processes to leverage both structural and semantic indicators of value when selecting high-potential research candidates for further exploration.

3.5. Generative interpretation and contextualization (GIC)

Building upon the semantically re-ranked candidate pairs from CREF, the Generative Interpretation and Contextualization (GIC) module aims to transform high-value predictions into interpretable and actionable research insights. This stage represents the final step in the knowledge service pipeline, where machine-driven reasoning evolves into human-facing inspiration, guiding new research avenues.

For each of the top-M candidate pairs in C_{CREF}, the GIC module generates a structured output that includes (i) the keyword pair (k_i, k_j), (ii) the value scores and justifications $S_f(k_i, k_j)$ from the CREF process, and (iii) the textual context $C(k_i, k_j)$ used for evaluation. The LLM is prompted to generate a detailed explanation that provides a semantic interpretation of the keyword connection and suggests potential research directions.

The prompt for the LLM in this module is designed as follows:

Prompt Example for GIC

> You are an expert in [domain] research, specializing in [specific field or topic]. Given the following pair of keywords, (k_i, k_j), please provide the following:
> 1. A brief definition of each keyword.
> 2. A rationale for why these two keywords are connected in the context of recent research.
> 3. Three potential research questions or hypotheses that explore the intersection of these two keywords.
> 4. Any additional insights or research directions that could arise from this pair.
> Please ensure the explanations are clear, relevant, and actionable for guiding further scientific exploration.

The output narrative consists of several components: - The LLM first defines k_i and k_j using domain-appropriate language, ensuring conceptual clarity. - It then provides a rationale for the connection between these two keywords, referencing mechanisms, theories, or technical links suggested by the context. - A restatement of the novelty and value dimensions is presented, based on the CREF evaluation, highlighting the significance of this research direction. - Finally, the LLM proposes three research questions $\mathcal{Q}_{ij} = \{q_1, q_2, q_3\}$ and hypotheses $\mathcal{H}_{ij}$ that could serve as initial starting points for further exploration. Optionally, broader future directions and related concepts may also be suggested.

The complete output is a human-readable "insight report", $\mathcal{R}(k_i, k_j)$, which encapsulates both the scientific relevance of the keywords and speculative creativity, ultimately supporting early-stage ideation and hypothesis formation in scientific research.

3.6. Interactive personalization and human-in-the-loop knowledge services

While the preceding modules operate in a largely automated pipeline, the framework is designed to support interactive, human-in-the-loop (HITL) workflows that enhance usability and personalization. In practical deployment, users may iteratively engage with the system to refine predictions, adjust evaluation criteria, and steer idea generation based on their evolving research interests.

In the CREF stage, researchers may provide feedback on the value assessments–e.g., expressing preference for higher novelty over interdisciplinarity–which can be used to reweight the aggregation function or modify the LLM prompt. Similarly, in the GIC stage, the insight reports can serve as seeds for interactive dialogue. A researcher may ask for clarification, request expansion on a particular mechanism, or prompt the LLM to suggest complementary concepts or experimental directions.

Such interactions not only improve the interpretability and relevance of the output but also enable dynamic adaptation to domain-specific needs. Over time, preferences inferred from user interaction history may be used to construct personalized value profiles or prompt templates, making the system increasingly aligned with individual discovery patterns.

Although a full user-adaptive implementation is beyond the scope of this paper, the proposed architecture is designed to accommodate such extensions, offering a pathway from one-shot recommendation toward adaptive, conversational knowledge services.

The training algorithm of GNN–LLM synergistic framework for scientific knowledge discovery is presented in Algorithm 2.

4. Experiments

This section presents the experimental evaluation of the GNN-LLM framework. The evaluation systematically assesses the core components: the performance of SE-TGN in identifying candidate keyword associations, the impact of the LLM-based CREF in refining these candidates, and the capabilities of the GIC module in generating human-readable insights and research prompts. Evaluations are performed on CS and Med corpora to test domain robustness. The subsequent subsections describe the datasets, evaluation metrics, and implementation details, followed by a presentation and discussion of quantitative and qualitative results.

4.1. Datasets

Two scientific corpora from the Web of Science were selected for evaluation: one from the domain of computer science (CS) and the other from biomedical research (Med). Each dataset spans publications from 2000 to 2020, with the period 2000–2018 designated for training, 2019 for validation, and 2020 for testing. Each document is annotated with a title, abstract, and a set of standardized keywords. Co-occurrence events are generated by enumerating all unordered keyword pairs within a paper, forming the temporal graph input. Table 1 summarizes key statistics across the three data splits. The CS corpus exhibits a larger vocabulary and higher co-occurrence volume, while the Med dataset shows greater sparsity and terminological variability.

Temporal dynamics in document counts, co-occurrence frequencies, and active vocabulary size are illustrated in Fig. 2. The Med dataset shows a sharper rise in recent years across all metrics, suggesting rapid expansion in biomedical literature and concept emergence, whereas the CS corpus demonstrates relatively steady growth.

Algorithm 2 GNN-LLM synergistic framework for scientific knowledge discovery.

Require: Scientific corpus $\mathcal{P}$, pretrained encoder f_{enc}
Ensure: Top-M keyword pairs with explanations and research ideas
Stage 1: Data Representation
for all paper $p_i \in \mathcal{P}$ **do**
  Extract keywords K_i and text $Text_i$ from p_i
  Compute semantic embedding $\mathbf{m}_{\text{pid}_i} \leftarrow f_{\text{enc}}(Text_i)$
end for
Construct keyword co-occurrence stream $\mathcal{S} = \{(u, v, \text{pid}, \tau)\}$
Stage 2: Candidate Generation via SE-TGN
Invoke Algorithm 1 on $\mathcal{S}$ to get ranked candidates
$C_{\text{EAPM}} \leftarrow \{((k_i, k_j), s_{ij}^{\text{GNN}})\}$
Stage 3: Semantic Re-ranking via CREF
for all $(k_i, k_j) \in C_{\text{EAPM}}$ **do**
  Retrieve top-k papers related to k_i and k_j
  Form context $C(k_i, k_j)$ using titles and abstracts
  Query LLM to score dimensions: Novelty, Impact, Plausibility, Interdisciplinarity
  Aggregate scores into S_{ij}^{LLM}
end for
Sort C_{EAPM} by S_{ij}^{LLM} to obtain C_{CREF}
Stage 4: Research Idea Generation via GIC
for all $(k_i, k_j) \in$ top-M pairs from C_{CREF} **do**
  Prepare LLM prompt using (k_i, k_j), S_{ij}^{LLM}, and $C(k_i, k_j)$
  Generate keyword definitions, rationale, research questions, hypotheses, etc.
end for
return Final enriched list of research points with ideas and explanations

Fig. 3 further visualizes semantic diversity using t-SNE projections of 384-dimensional sentence embeddings derived from paper titles and abstracts (encoded via the 'all-MiniLM-L6-v2' model). The clustering patterns highlight thematic distinctions and domain-specific distributional structures across the two corpora.

4.2. Evaluation metrics

To assess the performance of the GNN-based Emerging Association Prediction Module (EAPM) and subsequently the impact of LLM-driven refinements, we employ a set of standard evaluation metrics focused on ranking quality and top-K performance. Let $\mathcal{L}^+$ denote the set of true positive (emerging) keyword co-occurrences in the respective evaluation period (validation or test year), and $\mathcal{L}^-$ denote the set of true negative (non-occurring or non-novel) pairs.

4.2.1. AUC (Area under ROC curve)

AUC measures the probability that a randomly chosen positive instance from $\mathcal{L}^+$ is ranked higher by the model than a randomly chosen negative instance from $\mathcal{L}^-$. It provides a threshold-independent summary of overall ranking performance, with higher values indicating better discriminative ability.

Table 1
Corpus statistics of CS and Med datasets.

Dataset	Split	Docs	Events	Keywords	Period
CS	Train	37,922	216,250	54,423	2000–2018
	Valid	3686	22,776	8339	2019
	Test	1939	12,699	4997	2020
Med	Train	38,447	137,959	34,720	2000–2018
	Valid	3513	18,974	6735	2019
	Test	4272	12,874	4171	2020

4.2.2. AP (Average precision)

Average Precision (AP) summarizes the precision-recall curve into a single score. It is calculated as the weighted sum of precisions at each threshold where a relevant item is retrieved, with weights being the increase in recall. AP is particularly informative for tasks with imbalanced class distributions, such as link prediction.

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n, \tag{6}$$

where P_n and R_n are the precision and recall at the n-th item in the ranked list of all predictions.

4.2.3. P@K (Precision@K)

P@K evaluates the proportion of true positive links among the top-K predicted keyword co-occurrences. This metric is crucial for assessing the practical utility of the system in providing a manageable list of high-quality recommendations to researchers. It is defined as:

$$\text{P@K} = \frac{|\{\text{Top-K predicted links}\} \cap \mathcal{L}^+|}{K}. \tag{7}$$

In this study, we primarily report P@K for K ∈ {10, 50, 100, 500, 1000, 2000}.

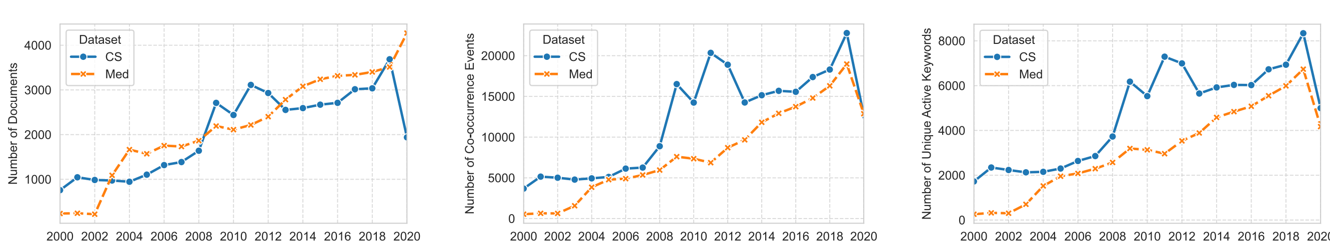
4.2.4. NDCG@K (Normalized discounted cumulative gain at K)

Normalized Discounted Cumulative Gain at K (NDCG@K) is a rank-aware metric that evaluates the quality of the top-K predictions by assigning higher scores to relevant items ranked earlier. The Discounted Cumulative Gain (DCG@K) is first calculated as:

$$\text{DCG@K} = \sum_{i=1}^{K} \frac{rel_i}{\log_2(i+1)}, \tag{8}$$

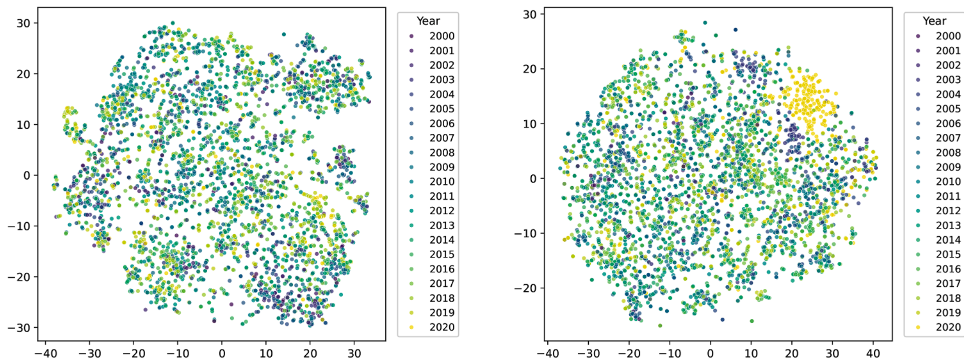
where $rel_i = 1$ if the item at rank i in the predicted list is a true positive (i.e., in $\mathcal{L}^+$), and $rel_i = 0$ otherwise. NDCG@K is then obtained by normalizing DCG@K with the Ideal DCG@K (IDCG@K), which is the DCG score of a perfect ranking:

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}. \tag{9}$$



(a) Documents per year (b) Co-occurrence events (c) Active keywords

Fig. 2. Temporal trends in publication volume, co-occurrence frequency, and vocabulary growth for CS and Med.



(a) CS Embeddings (b) Med Embeddings

Fig. 3. The t-SNE projection of semantic embeddings from sampled documents.

Table 2
Transposed GNN link prediction results on CS dataset.

Metric	Static GNN			Cumulative GNN			SE-TGN (Ours)		
	GCN	SAGE	GAT	GCN	SAGE	GAT	GCN	SAGE	GAT
AUC	0.662	0.442	0.637	0.570	0.544	0.572	0.623	0.617	**0.719**
AP	0.292	0.145	0.321	0.316	0.113	0.201	0.659	0.669	**0.670**
Prec@10	**1.000**	0.800	**1.000**	**1.000**	0.000	**1.000**	**1.000**	**1.000**	**1.000**
Prec@50	**1.000**	0.580	0.940	**1.000**	0.100	0.920	**1.000**	**1.000**	**1.000**
Prec@100	**1.000**	0.590	0.920	0.990	0.120	0.940	**1.000**	0.990	**1.000**
Prec@500	0.970	0.524	0.890	0.944	0.144	0.772	**0.996**	0.988	0.992
Prec@1k	0.889	0.523	0.826	0.876	0.145	0.691	0.991	0.989	**0.993**
Prec@2k	0.716	0.491	0.670	0.764	0.155	0.544	0.961	**0.970**	0.961
NDCG@10	0.432	0.299	0.372	0.482	0.000	0.450	0.734	**0.902**	0.889
NDCG@50	0.494	0.282	0.432	0.517	0.037	0.452	**0.816**	0.779	0.798
NDCG@100	0.542	0.306	0.466	0.616	0.059	0.576	**0.818**	0.519	0.551
NDCG@500	0.781	0.425	0.707	0.654	0.112	0.679	**0.815**	0.641	0.620
NDCG@1k	0.794	0.465	0.736	0.646	0.125	0.658	**0.818**	0.705	0.687
NDCG@2k	0.701	0.468	0.654	0.752	0.142	0.553	0.720	**0.759**	0.746

A higher NDCG@K indicates a better ranking quality, where relevant items are positioned higher. We report NDCG@K for K ∈ {10, 50, 100, 500, 1000, 2000}.

Our primary analysis emphasizes P@K and NDCG@K, complemented by the global ranking metrics AUC and AP. This focus is motivated by the practical application of discovering novel research points. Researchers typically require a manageable list of high-quality, promising leads. High P@K directly reflects the system's utility in providing such actionable starting points. However, as P@K can be high for several methods (especially at small K), NDCG@K is introduced as a more discriminative metric. It not only considers correctness but also the rank position of correct predictions within the top-K, offering a nuanced evaluation of how well the most relevant items are prioritized. While Recall@K (R@K) and F1@K are computed and logged for completeness, they are less emphasized in our main comparative tables because the total number of true new links in a scientific domain over a year can be vast, leading to inevitably low R@K and F1@K for practical values of K, making them less effective for distinguishing the top-end performance of different methods in this context.

4.3. Implementation details

All GNN models are implemented using PyTorch. The default embedding dimension is 64, and hidden channels are also set to 64. For snapshot models, we use a learning rate of 0.001 and train for up to 100 epochs, with dropout of 0.3 and a binary cross-entropy (BCE) loss using a positive weight of 1.0. For SE-TGN models, we adopt a mini-batch size of 256, a learning rate of 1×10^{-5}, and a positive BCE weight of 10.0. During training, negative samples are generated via random sampling; for each positive keyword pair in a batch, we randomly sample a pair of keywords that is not observed to co-occur in the training data as a negative instance. Early stopping is applied with a patience of 10 epochs based on validation AP.

Paper data is embedded using the pretrained 'all-MiniLM-L6-v2' encoder, producing 384-dimensional semantic vectors from concatenated titles and abstracts. These embeddings serve as dynamic edge features for SE-TGN.

For the LLM component, we use GPT-4o as the default instruction-tuned model to perform semantic value assessment and research idea generation, accessed via the OpenAI API. A temperature setting of 0.7 was used for all generation tasks to ensure a balance between creativity and factual grounding. The top-N = 1000 GNN predictions are passed to the LLM for re-ranking, and the top-M = 30 pairs are selected for final insight generation.

4.4. Results of GNN-based candidate discovery

This section presents the performance of GNN-based models in candidate discovery, a key stage in our framework responsible for identifying structurally and temporally promising concept associations. We evaluate a diverse set of GNN architectures on both the CS and Med datasets, including static models trained on cumulative or annual graph snapshots, and our proposed SE-TGN, which operates on continuous temporal event streams while incorporating document-level semantic signals. Together, these results help isolate the impact of temporal modeling and semantic input on candidate discovery performance.

Table 3
Transposed GNN link prediction results on med dataset.

Metric	Static GNN			Cumulative GNN			SE-TGN (Ours)		
	GCN	SAGE	GAT	GCN	SAGE	GAT	GCN	SAGE	GAT
AUC	0.665	0.372	0.665	0.551	0.499	0.581	0.670	0.642	**0.695**
AP	0.250	0.070	0.250	0.222	0.142	0.183	0.710	0.713	**0.718**
Prec@10	**1.000**	0.000	**1.000**	**1.000**	**1.000**	0.700	**1.000**	**1.000**	0.900
Prec@50	0.940	0.060	0.940	0.940	0.880	0.730	0.960	**1.000**	0.980
Prec@100	0.940	0.068	0.940	0.720	0.480	0.631	0.980	**1.000**	0.990
Prec@500	0.812	0.098	0.812	0.610	0.360	0.512	0.990	**0.994**	0.992
Prec@1k	0.709	0.105	0.709	0.520	0.265	0.435	0.989	**0.991**	0.987
Prec@2k	0.526	0.117	0.526	0.395	0.220	0.327	**0.986**	0.985	0.983
NDCG@10	0.031	0.000	0.031	0.027	0.021	0.024	0.814	0.781	**0.840**
NDCG@50	0.061	0.021	0.061	0.043	0.032	0.038	**0.845**	0.498	0.775
NDCG@100	0.085	0.025	0.085	0.057	0.036	0.050	0.694	0.582	**0.750**
NDCG@500	0.167	0.058	0.167	0.110	0.061	0.088	0.779	0.691	**0.794**
NDCG@1k	0.217	0.070	0.217	0.136	0.078	0.104	0.440	**0.722**	0.424
NDCG@2k	0.261	0.136	0.240	0.278	0.229	0.193	0.584	**0.728**	0.583

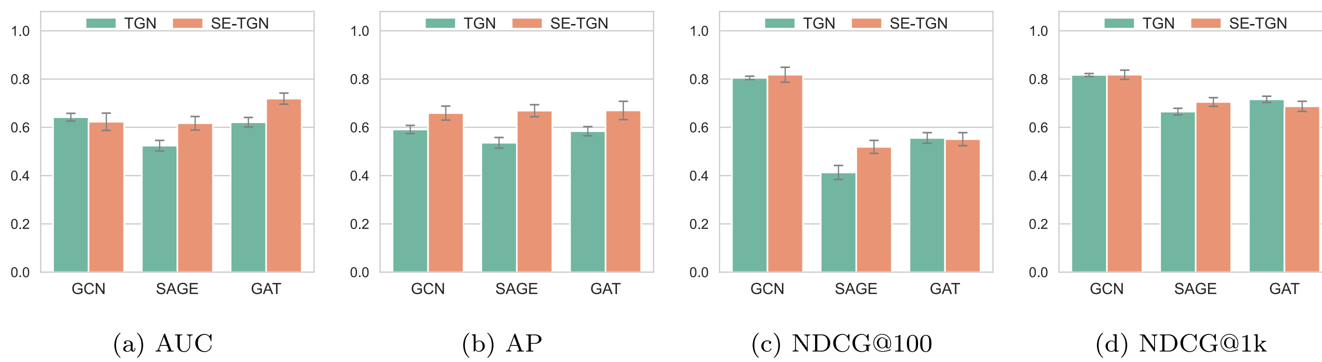


Fig. 4. Ablation study on the CS dataset: TGN vs. SE-TGN across GCN, SAGE, and GAT backbones.

Tables 2 and 3 summarize the link prediction results across a wide range of ranking metrics. Overall, SE-TGN models consistently outperform their static and snapshot-based counterparts on both datasets. On the CS dataset, SE-TGN with GAT backbone achieves the highest AUC (0.719) and AP (0.670), while all SE-TGN configurations attain perfect or near-perfect Precision@10 and NDCG@100. Moreover, SE-TGN (GCN) and (SAGE) demonstrate strong performance at deeper ranks (e.g., NDCG@2k), indicating robust retrieval depth. A similar pattern is observed on the Med dataset, where SE-TGN (GAT) again leads in AUC (0.695) and AP (0.718), and SE-TGN (SAGE) performs best on broader ranking metrics like NDCG@1k and NDCG@2k. These results highlight the benefit of jointly modeling temporal dynamics and semantic features, particularly for longer-range discovery.

To isolate the contribution of semantic features, we conduct an ablation study comparing TGN models with and without semantic enhancement. Figs. 4 and 5 illustrate the results across five metrics (AUC, AP, P@1k, NDCG@100, NDCG@1k) for three model backbones.

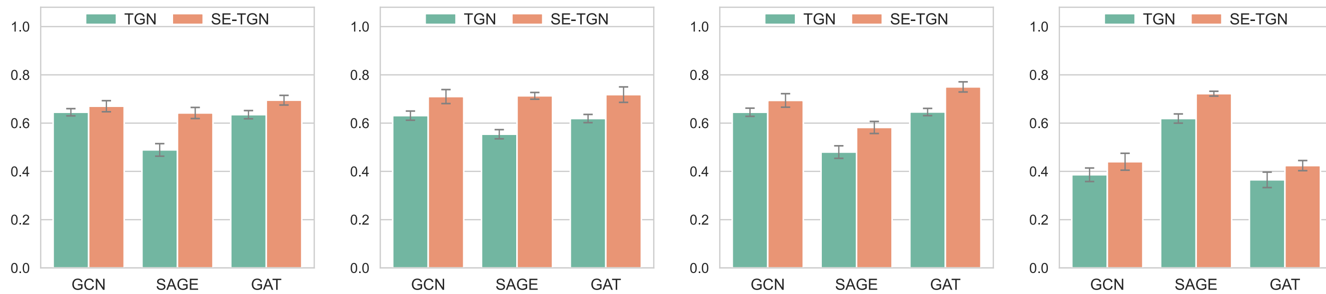
Across both datasets, semantic enhancement yields consistent gains, especially for GAT-based models where attention mechanisms can leverage the richer input signal. Improvements are particularly pronounced in AP and NDCG@100, which reflect the model's ability to prioritize high-quality candidates. For SAGE and GCN, the improvements are generally positive but sometimes modest–likely because these architectures rely more on local aggregation, which may already capture dominant patterns in denser graphs.

We also observe that SE-TGN variants tend to have narrower error bars, suggesting reduced variance and more stable convergence during training. This supports our hypothesis that semantic features not only improve accuracy, but also enhance model robustness and generalization.

4.5. Impact of semantic re-ranking via CREF

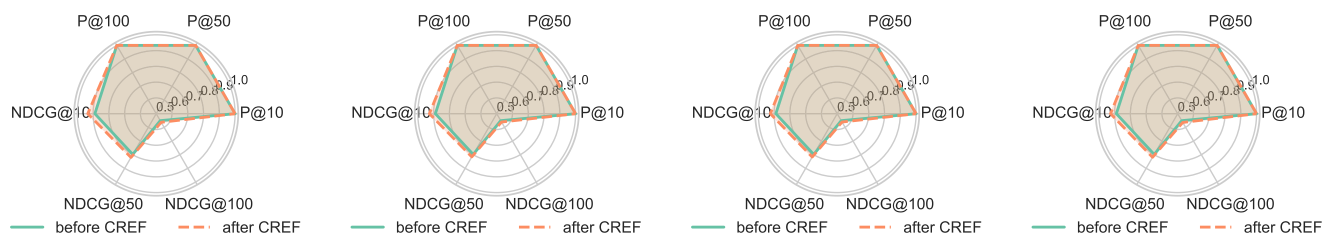
Following the establishment of the structural soundness of SE-TGN-generated candidates, the effectiveness of the Contextual Re-ranking and Evaluation Framework (CREF) in refining the top-N candidates based on language model (LLM)-guided semantic assessment is now evaluated. While SE-TGN scores provide a statistical likelihood of future co-occurrence, they do not account for key dimensions such as novelty, interdisciplinarity, or the broader research potential of the candidate pairs. CREF addresses this gap by leveraging the capabilities of a large language model (here, GPT-4o) to assess each candidate keyword pair in the context of surrounding paper content.

The impact of semantic re-ranking is quantified by comparing ranking metrics of the original SE-TGN output with those refined by CREF, across various (M,N) configurations. Figs. 6 and 7 provide a visual comparison of the results for the CS and Med datasets, respectively. The data consistently demonstrate that CREF enhances ranking quality across the board. For the CS dataset (illustrated in Fig. 6), while the SE-TGN GAT model already achieves perfect Precision@k scores (P@10, P@50, P@100 = 1.000), CREF maintains these maximum precision scores across all configurations. More notably, CREF produces substantial improvements in ranking measures like NDCG. For instance, in the (M = 1000, N = 100) setting, NDCG@10 improves from 0.889 (SE-TGN GAT) to 0.934, and NDCG@100 increases from 0.551 to 0.563. These improvements are a strong indication that CREF significantly enhances the quality of ranked candidates, even when the initial precision is already high. The Med dataset (Fig. 7) also demonstrates the effectiveness of CREF, particularly in improving Precision@k scores. For example, P@10 increases from 0.900 to 0.934, and P@50 rises from 0.980 to 1.000 in the (M = 1000, N = 100) setting. These improvements reflect



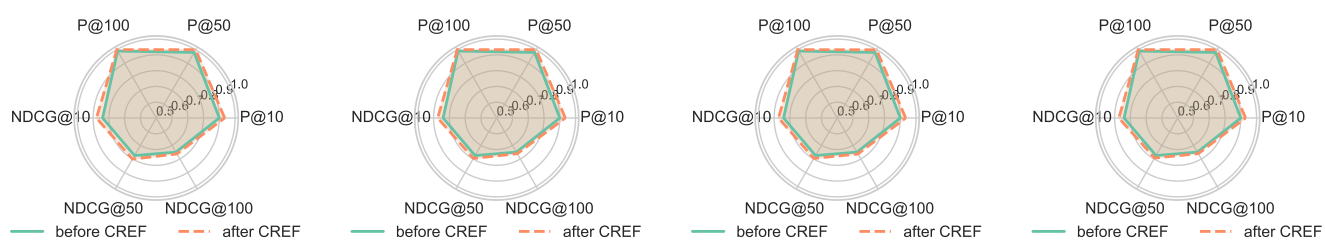
(a) AUC (b) AP (c) NDCG@100 (d) NDCG@1k

Fig. 5. Ablation study on the Med dataset: TGN vs. SE-TGN across GCN, SAGE, and GAT backbones.



(a) M=1000, N=100 (b) M=500, N=100 (c) M=2000, N=500 (d) M=1000, N=500

Fig. 6. Radar charts comparing performance before and after CREF for different (M, N) configurations on the CS dataset.



(a) M=1000, N=100 (b) M=500, N=100 (c) M=2000, N=500 (d) M=1000, N=500

Fig. 7. Radar charts comparing performance before and after CREF for different (M, N) configurations on the Med dataset.

CREF's ability to refine the list of predictions, elevating candidates with higher scientific value.

The observed gains–especially in NDCG for the CS dataset and in Precision@k for the Med dataset–suggest that the LLM can refine the GNN-generated output by better prioritizing those links with higher research potential. This refinement is crucial, even when initial precision is high (as seen in the CS dataset with P@k scores of 1.000), because CREF helps to highlight the most impactful candidates among many structurally plausible ones. These are candidates that may have been under-ranked in terms of overall scientific merit by the GNN alone, primarily due to its reliance on structural patterns derived from internal graph data.

In essence, CREF serves as a complementary layer to the pattern-recognition capabilities of SE-TGN, which primarily focuses on structural patterns and co-occurrence patterns within the knowledge graph. By integrating a semantic assessment driven by an LLM, which utilizes open-domain literature (external information), CREF introduces an orthogonal layer of evaluation. This dual-perspective optimization ensures that the final rankings prioritize candidates that are not only structurally valid (based on internal data patterns) but also semantically rich in terms of research potential, novelty, and impact. The integration of this semantic evaluation step ensures that the final list of candidates is both statistically sound and scientifically insightful, paving the way for the final research insight generation stage.

4.6. Interpretability and inspiration from GIC module

The Generative Interpretation and Contextualization (GIC) module plays a crucial role in transforming the structural connections identified by the GNN model into clear, actionable, and semantically enriched research directions. By bridging the gap between raw graph-based data and human-interpretable knowledge, GIC equips researchers with insights that are immediately applicable in early-stage ideation or interdisciplinary exploration. It turns abstract graph relationships into tangible research opportunities, making it an invaluable tool for guiding scientific discovery.

To demonstrate the impact of GIC, we present a detailed case study involving the pair of keywords “evolutionary algorithms” and “multi-objective optimization”. This case study highlights how GIC can provide researchers with the necessary context, clarify the relationship between these keywords, and generate actionable research directions.

This case study clearly demonstrates the potential of the GIC module to transform keyword relationships into meaningful, research-relevant insights. The generated definitions, rationale, and hypotheses serve as a solid foundation for researchers to frame their investigations, ensuring they tackle not only theoretically interesting but also practically impactful problems.

The ability of GIC to generate research questions and offer actionable directions based on structural data significantly enhances its value in the research pipeline. By providing both context and clarity, it empowers researchers to delve deeper into the relationships between concepts, encouraging innovative thinking that transcends traditional boundaries and fostering interdisciplinary research. Furthermore, the insights and research directions proposed by GIC provide a concrete foundation for developing new theories or refining existing models, underscoring the essential role this module plays in advancing scientific knowledge discovery.

A Case Study for Knowledge Service: "Evolutionary Algorithms" and "Multi-objective Optimization"

1. **Brief Definitions of Each Keyword:**
   - *Evolutionary Algorithms (EAs):* Optimization techniques inspired by natural evolution, such as Genetic Algorithms (GAs), that iteratively evolve a population of solutions through selection.
   - *Multi-objective Optimization (MOO):* Optimization problems involving multiple conflicting objectives. Solutions are evaluated based on Pareto optimality, where no objective can be improved without worsening another.
2. **Connection:** EAs are well-suited for MOO as they handle complex, high-dimensional search spaces. Multi-objective optimization often requires exploring a diverse set of solutions, which EAs naturally facilitate through population-based search and Pareto dominance. Recent research has focused on improving diversity and scalability in these methods.
3. **Research Questions:**
   - *How can the combination of evolutionary algorithms and machine learning techniques improve the efficiency of solving large-scale multi-objective optimization problems?*
   - *What methods can be employed to improve diversity maintenance in evolutionary algorithms for multi-objective optimization in highly dynamic or uncertain environments?*
   - *How can multi-objective evolutionary algorithms be adapted to optimize non-differentiable, multi-modal functions encountered in engineering design problems?*
4. **Additional Insights or Research Directions:**
   - *Hybridization:* Combining EAs with other methods (e.g., particle swarm optimiza- tion or deep learning) may enhance optimization performance.
   - *Surrogate Models:* Using surrogate models can reduce computational costs in expensive objective function evaluations.
   - *Automated Parameter Tuning:* Methods like meta-learning can automate the tuning of EA parameters for better performance.
   - *Applications:* EAs and MOO are valuable in fields like sustainability, resource allocation, and healthcare, where multiple objectives need to be balanced.

In summary, the GIC module is a powerful tool for enhancing the knowledge discovery process. By converting abstract keyword associations into comprehensive, actionable research prompts, it bridges the gap between raw structural data and human-centric scientific inquiry. This integration of graph-based structural data with semantic interpretation allows researchers to understand connections between concepts and explore groundbreaking research directions that may have been previously overlooked.

5. Discussion

The proposed framework illustrates the benefit of combining temporal graph modeling with semantic reasoning for scientific knowledge discovery. The SE-TGN effectively captures the dynamic co-evolution of scientific concepts, augmented by paper-level semantic embeddings. This enables the identification of structurally promising, yet potentially novel, keyword associations. However, structural plausibility alone does not guarantee research value. The subsequent LLM-based modules–CREF and GIC–address this gap by introducing value-aware filtering and human-readable explanation, effectively transforming candidate links into plausible research directions.

Several limitations are acknowledged. The current implementation focuses on two domains and represents scientific ideas through keyword pairs, which may omit richer conceptual structures. The subjective nature of evaluating research novelty and insight also presents challenges, as does the reliance on generic prompting strategies in the LLM modules. Furthermore, the pipeline operates in a unidirectional fashion, limiting feedback between components and user influence during the discovery process.

6. Conclusion

This work presents a modular GNN-LLM framework that integrates temporal graph modeling with semantic evaluation and research idea generation to assist in the discovery of novel scientific insights. By combining the structural strengths of SE-TGN with the reasoning capabilities of LLMs, the framework identifies potentially meaningful concept associations and translates them into interpretable, actionable research directions.

Future work will explore broader domain coverage, richer semantic structures beyond keyword co-occurrence, and more interactive, adaptive mechanisms for HITL discovery. Further alignment between graph-based prediction and language-based evaluation–such as co-optimization or iterative feedback–may enhance the coherence and creativity of machine-assisted scientific exploration.

CRediT authorship contribution statement

Qingqing Wang: Writing – review & editing, Resources, Data curation, Conceptualization; **Derui Lyu:** Visualization, Software; **Qiuju Chen:** Writing – original draft, Supervision, Investigation.

Data availability

The authors do not have permission to share data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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